Mouhamadou Lamine SOW

Computer Science Engineering Student — Operations Research and Optimization Internship

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Profile

Final-year engineering student at Sup Galilée, focused on **operations research and decision support**: MILP modeling, metaheuristics and applied machine learning, put into practice on the ROADEF/EURO 2020 challenge and at **Sonatel (Orange Group)**. Available for a **six-month final-year internship from March 2027**.

Professional Experience

Machine Learning / Data Science Intern May 2026 – Jul 2026
Sonatel (Orange Group) Dakar, Senegal

- Designed an XGBoost model predicting network risks 6 months ahead: **88 % fewer false negatives** than the existing business rule (PR-AUC 0.81 vs. 0.28).
- Aggregated risk by area over **4 horizons from 3 to 12 months** to **prioritize field interventions**; calibrated, explainable probabilities (SHAP).
- Built, with a teammate, an **8-year fiber network history** (2018–2026); engineered temporal and geographic features with leakage-free chronological evaluation.

Job-Shadowing Internship — Digital Technology and DevOps Jul 2024 – Sep 2024
Sonatel (Orange Group) Dakar, Senegal

- Wrote a literature review on **intelligent DevOps automation**; introduction to Angular, LoopBack and MongoDB.

Projects

ROADEF/EURO 2020 — RTE Maintenance Scheduling 2026

Tools: Python, Gurobi, MILP, simulated annealing, local search

- Built a Gurobi MILP model strengthened with valid inequalities: **8 proven optima out of 15 instances**.
- Designed a greedy repair heuristic followed by simulated annealing: **100 % feasible solutions**, verified with the official checker.
- Cut the mean gap to the best known solution to **0.52 %** on 12 test instances (vs. 2.83 % for the greedy baseline).

Ring-Star — Transport Network Optimization Academic project

Tools: Python, PuLP, Gurobi, CPLEX, CBC

- Formulated a ring-star network design problem as a **MILP** (station selection, ring routing, user assignment), solved with Gurobi, CPLEX and CBC.
- Developed a heuristic and a local search: **19 % lower cost** than the greedy approach, in under one second.

Education

Engineering Degree in Computer Science — Master's level, expected 2027 Sep 2024 – Present
Sup Galilée — Université Sorbonne Paris Nord Villetaneuse, France

Key coursework: operations research, combinatorial optimization, algorithms, machine learning, NLP.

Integrated Preparatory Program (CPI) — MPSI Sep 2022 – May 2024
ESIGELEC–ESMT Dakar, Senegal

Mathematics, physics and engineering sciences.

Skills

Operations Research	MILP, combinatorial optimization, metaheuristics, local search
Solvers	Gurobi, CPLEX, CBC, PuLP
Programming & Data	Python, Julia, SQL; pandas, NumPy, SciPy
Machine Learning	scikit-learn, PyTorch; imbalanced classification, calibration, SHAP
Tools	Git, Linux, pytest, Jupyter

Languages & Certification

French: native **English:** B2 — TOEIC 802

IBM AI Engineering Professional Certificate — IBM / Coursera, in progress (2026).